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DESCENDING MOTOR PATHWAYS

----- Active space --

Brain → Spinal cord → Skeletal muscle.

Efferent/descending pathway : Top down approach.

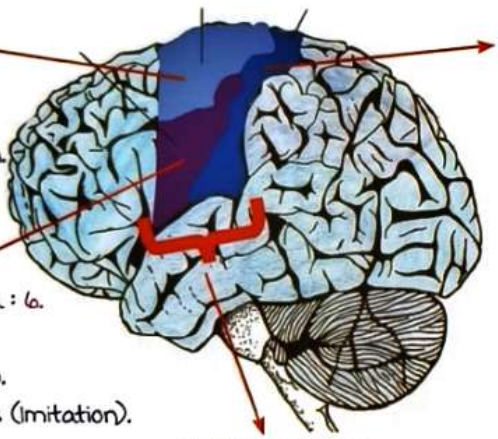
Cortex, Brainstem & Spinal cord Pathways

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Neurons involved in cortex & brain stem : Upper motor neurons.

CORTEX :

Motor cortex :



2. Supplementary :
   - Brodmann area : 6.
   - Function :
     - Bimanual coordination.

1. Primary :
   - Brodmann area 4.
   - Function : Execution of movement.

3. Pre motor :
   - Brodmann area : 6.
   - Function :
     - Postural set up.
     - Mirror neurons (Imitation).

4. Motor cortex :
   - Precentral gyrus (frontal lobe).

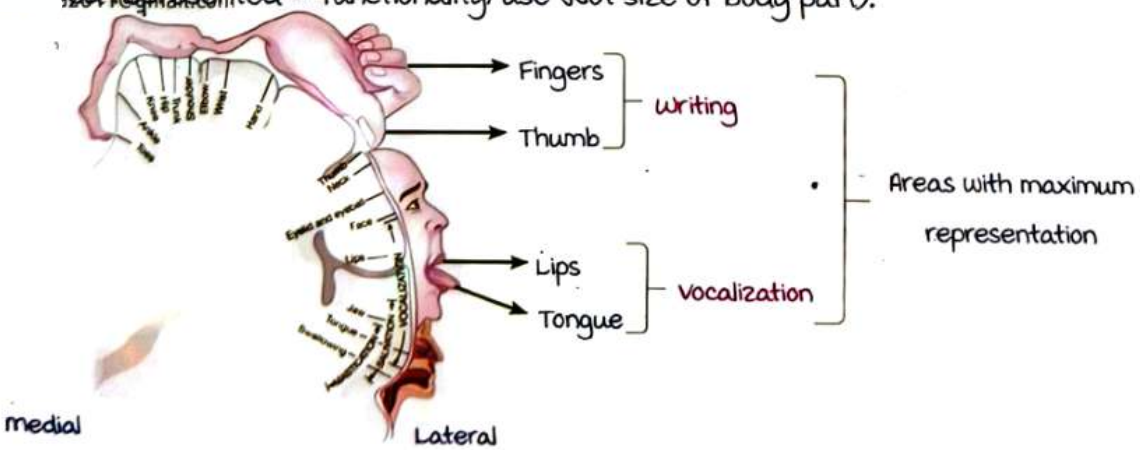
4. Cortical association areas :

Function : Idea generation & initiation of movement.

Note : Sensory cortex : Brodmann areas 3, 1, 2.

Motor homunculus/penfield homunculus :

- Body part representation.
- Area represented ∝ functionality/use (Not size of body part).

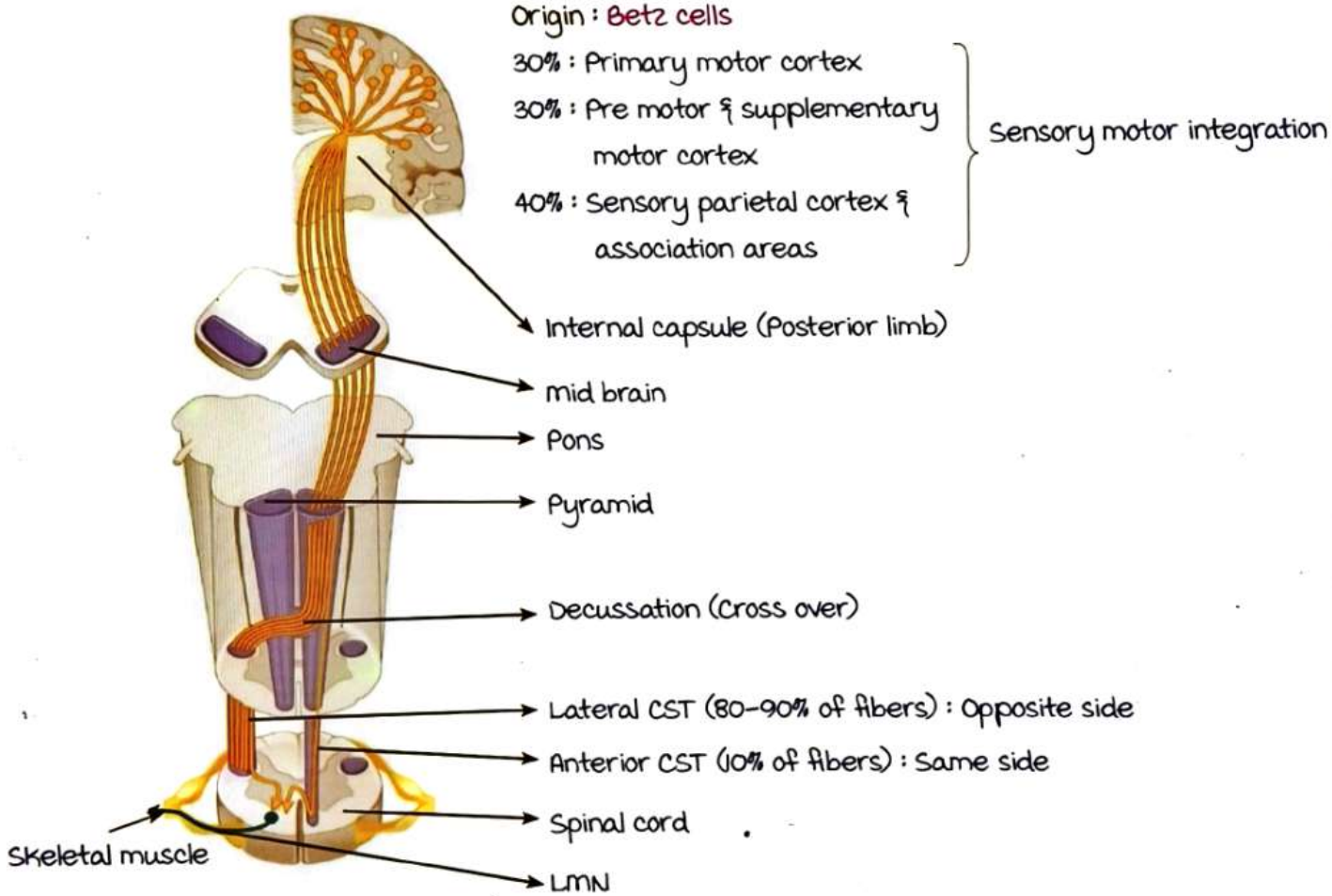


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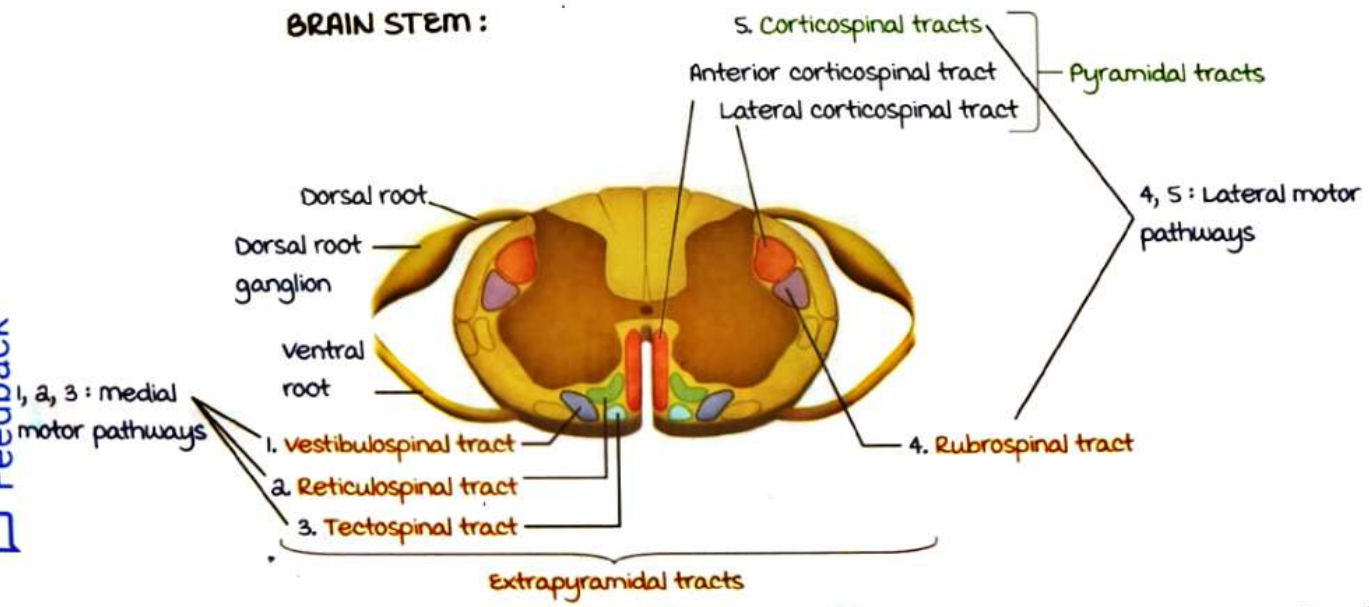
----- Active space -----

Corticospinal tract (CST)/pyramidal tract :

- Recently evolved.
- Contralateral pathway : Controls opposite side of body.
- Function :
  - Controls distal muscles.
  - Fine, skilled, voluntary motor movements (Eg : Writing).



BRAIN STEM :



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Extrapyramidal tracts :

----- Active space -----

1. Vestibulospinal tract
   - Medial → Controls neck musculature
   - Lateral → Maintains Posture
     - Muscles involved : Antigravity muscles
       - Paravertebral.
       - Proximal extensors.

2. Reticulospinal tract
   - Pontine : Excitatory → Maintains Posture
   - Medullary : Inhibitory → Maintains Posture

3. Tectospinal tract :

Origin : Superior colliculus.

Controls head & eye movements.

4. Rubrospinal tract :
   - Only extrapyramidal tract which is lateral.
   - Origin : Red nucleus.
   - Controls flexors of upper limb.

Lateral v/s medial motor pathways :

Features	Lateral	Medial
Tracts involved	Corticospinal, Rubrospinal	Vestibulospinal, Reticulospinal, Tectospinal
With respect to evolution	New	Old
Function	Fine, skilled, voluntary movements	Maintenance of posture
Type of muscle supplied	Flexors	Extensors (Antigravity muscles)

SPINAL CORD :

Final common pathway



Spinal Cord

LMNs (Lower motor neurons)

α motor neurons :
- Contracts muscle.
- Overactivity → spasticity

γ motor neurons :
- Regulates muscle tone.
- Overactivity → ↑ muscle tone.

Extrafusal fibres

Actin

Myosin

A M

muscle spindles

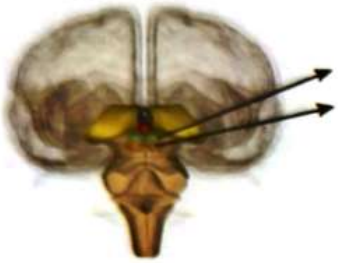
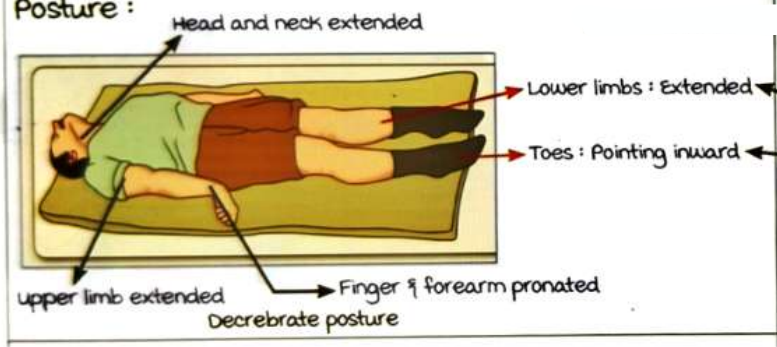
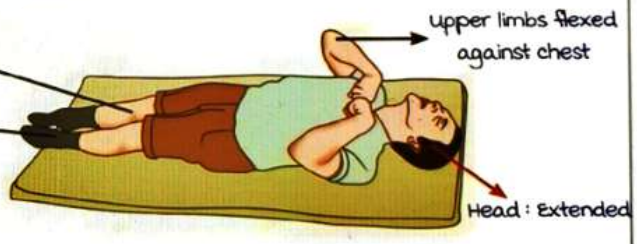
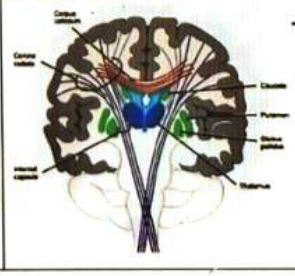
Intrafusal fibres :
- Maintains muscle tone.

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Disorders

Extensor rigidity	Flexor rigidity
Decerebrate rigidity :	**Decorticate rigidity**
Pathophysiology :	Pathophysiology
	Decortication (Cortex removed)
Transection b/w Superior colliculus & Inferior colliculus	Cortical inhibition lost ; Rubrospinal tract : Intact
↓ Cortical inhibition lost	↑ flexor muscle tone
↓ Dorsal sensory roots : Activated	↓ Decorticate rigidity
↓ ++ (Brainstem pathway) Reticulospinal tract	
↓ ++ γ - motor neurons	
↓ ++ ↑ extensor muscle tone	
↓ Decerebrate rigidity/ γ rigidity	
Posture :  Decerebrate posture	Posture :  Decorticate posture
Cause :  Uncal herniation : (D/t space occupying lesion) ↓ Decerebrate rigidity	Cause :  Thrombosis/hemorrhage of internal capsule ↓ Decorticate rigidity

Decerebellate rigidity :

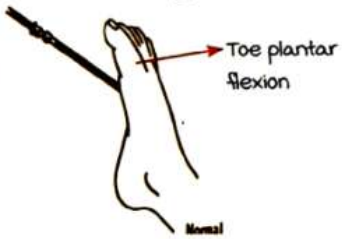
Anterior cerebellum removed —++→ motor neurons ↓ Exaggeration of extensor activity ↓ Decerebellate/α rigidity

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UMN V/S LMN LESIONS :

----- Active space -----

Signs	UMN lesions	LMN lesions
muscle atrophy	-	+++
muscle fasciculations	-	+++
muscle tone	↑↑	↓
weakness distribution	Regional	Segmental
Reflexes	↑↑ (hyper)	↓↓ (sluggish)
Babinski sign	+  Positive (+) Babinski sign (dorsiflexion of big toe)	-  Normal

Feedback